

11	THERMAL INSULATION		
	Supply, Installation of Insulation material. It shall be Class "O" type low fire propagation with closed cell elastomeric nitrile rubber having factory laminated glass reinforced minimum 12 microns pure aluminum foil with high low water vapour diffusion of 12,000 and low thermal conductivity. Material shall be elastomeric foam based on synthetic rubber conforming to BS 476part 6 for fire propagation test and Class 1 as per BS 476 part 7 for surface spread of flame test. Minimum 4" mm width glass reinforced high bonding Aluminium tape shall be applied on all transpose & longitudinal joints. All insulation joints (including Flange joints) to be sealed with 4" width 1mm Thk Self Adhesive tape. Density of Material shall be between 40-55 Kg/m3. Thermal conductivity shall not exceed 0.035 W/m/K at mean temperature of 0°C. Thickness of the insulation will be as specified for the individual application. Adhesive used for setting the insulation will be non-flammable, vapour proof adhesive low voc igbc compliant Supply of Adhesive will be in scope of HVAC contractor. Random materials test for supplied insulation to be conducted from Nabl certified labs for Class O & K value by the user. Minimum 1 day training to be given to the application team by manufacturer before work commencement for good workmanship and Fortnite visit to be conducted to check the application quality. Report to be submitted the user, pmc and consultant		
11.1	32 mm thick for Clean Room SA Duct	Sqm.	15000
11.2	25 mm thick for Clean Room SA Duct	Sqm.	23750
11.3	19 mm thick for Comfort SA Duct	Sqm.	8500
11.4	13 mm thick for Comfort RA Duct	Sqm.	5000
12	ACOUSTIC LINING		
	Supply, Installation and Testing of Accoustic lining with 10 mm thick Class 1 rating Open Cell nitrile rubber insulation material with density of 140 to 180 kg /m³ along with manufacturer recommended adhesive.	Sqm.	500
8	S.I.T.C. of Thermal insulation of CHW Pipings with low fire propagation Class O & UL94(VO,HB Passed) FM Approved Closed Cell Nitrile Rubber having active anti microbial Microban properties with K value of 0.035 w/m.k at 0 Deg C Temp, and having µ value 7,000. Anti Microbial Additive to be EPA Approved. Insulation covering to be 26G aluminium cladding, coated with a special UV protection. Adhesive to be used should be GRIHA Approved and specially formulated for Ducting Insulation. Good Quality 50 mm nitrile rubber tapes .Installation Should be as per manufacturers SOP. Price shall include all the accessories and PUF Block support of suitable density as per pipe and fixing arrangement with Structural support		
8.1	75 mm thick Insulation on 1200NB to 1000NB	Sq. Mtr.	1,976.00
8.2	51 mm thick Insulation on 950NB to 500NB	Sq. Mtr.	3,600.00
8.3	38 mm thick Insulation on 450NB to 80NB	Sq. Mtr.	14,800.00
8.4	19 mm thick Insulation on 65NB / 50NB / 40NB / 32NB / 25NB / 20NB / 15NB	Sq. Mtr.	7,400.00
6	Supply & Installation of underfloor insulation as per specification. Material shall be 19mm thick Class " 0 " Closed cell nitrile rubber elastomeric foam insulation laminated with chemically treated glass cloth of 7 mil / 0.18 mm thickness Thermal conductivity of insulation material shall not exceed 0.036 W/(m.K) at mean temperature of 24°C as per ASTM C 518 / ASTM C 177. The insulation material shall have water absorption of 0.2% by volume as per ASTM C 1763 ,shall have fire performance rating of 25/50 as per ASTM E 84. Density will be 40 to 60 Kg/m3 . • The insulation material shall have Microban** an anti-microbial product, which is EPA (Environmental Protection Agency), USA Approved, as an integral part of insulation that cannot be washed off or worn off.The insulation material shall give enhanced level of protection against harmful Microbes such as fungi and mold and should confirm to ASTM G 21, UL 181 & ASTM E 2180. Water Vapor Permeability of insulation material shall be minimum 0.05 Perm-inch. without any external barrier as per ASTM E 96 Procedure A. The adhesives for the application of insulation should be of principal manufacturer. Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation	Sqm	R/O
7	Supply & Installation of underdeck insulation as per specification. Material shall be 19mm thick Class " 0 " Closed cell nitrile rubber elastomeric foam insulation laminated with chemically treated glass cloth of 7 mil / 0.18 mm thickness Thermal conductivity of insulation material shall not exceed 0.036 W/(m.K) at mean temperature of 24°C as per ASTM C 518 / ASTM C 177. The insulation material shall have water absorption of 0.2% by volume as per ASTM C 1763 ,shall have fire performance rating of 25/50 as per ASTM E 84. Density will be 40 to 60 Kg/m3 . • The insulation material shall have Microban** an anti-microbial product, which is EPA (Environmental Protection Agency), USA Approved, as an integral part of insulation that cannot be washed off or worn off.The insulation material shall give enhanced level of protection against harmful Microbes such as fungi and mold and should confirm to ASTM G 21, UL 181 & ASTM E 2180. Water Vapor Permeability of insulation material shall be minimum 0.05 Perm-inch. without any external barrier as per ASTM E 96 Procedure A. The adhesives for the application of insulation should be of principal manufacturer. Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation	Sqm	R/O

Plain NBR to Quote

compliance
complies for 40-60 kg/m3
complies (density will be 180-220 kg/m3)
complies
complies
complies